ALIKHAN KABDUALY

Building AI products at 16. From Kostanay, Kazakhstan.

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ABOUT

I'm Alikhan, a 16-year-old developer from Kostanay, Kazakhstan. I build AI-powered products — computer vision models, progressive web apps and Telegram bots — taking them from concept to deployment. I've competed in hackathons and pitch sessions across the Astana Hub ecosystem in Kostanay, Petropavlovsk and Almaty. In September 2026 our team took first place at the Qostanai AgroTech Hackathon with AgroVision AI, a weed map built from drone imagery, and in October we reached the final of the Qostanai AI Industry Hackathon with НарядAI, a repair work-order system for Kostanay Minerals. Right now I'm building a new project — no details yet.

PROJECTS AND VENUES

НарядAI

2026-10 · 6 days of building + work before the final

Repair work orders with an AI inspector for a mining and processing plant

Venue: Qostanai AI Industry Hackathon 2026 · case No. 1 · finalist

Role: Captain and lead developer — the architecture, the database and its access rules, the AI modules and the Laya fine-tune, the worker's native Android app

Stack: Next.js 16 · React 19 · TypeScript · Supabase · PostgreSQL RLS · PWA · Kotlin · Laya · Whisper · Qwen3 · DeepSeek · Telegram Bot API

- 85 / 100 points in selection, 4th place on the case
- 12 teams in the final out of 150+ applications
- 93% Laya category accuracy on 192 descriptions
- 6 taps to issue an order

AgroVision AI

2026-09 · 2 days + post-event work

A weed map and spot-spraying plan from drone imagery

Venue: Qostanai AgroTech Hackathon 2026 · case No. 1 · 1st place

Role: Development and ML — the detector and classifier, the agronomy rules, georeferencing and task export for machinery, the server, the agronomist's offline app and the dashboard

Stack: Python · PyTorch · YOLOv8s · EfficientNet-B0 · FastAPI · Streamlit · PWA · GeoJSON · ISO 11783-10 · Docker

- 0,77 detector mAP@50 on an independent test, 381 images
- 26 weed species and 3 stages in the classifier
- 98,4% species accuracy on 1,591 reference photos
- 500 000 ₸ prize for first place

Jasyl

2026-08 · 2 days

AI monitoring of Kostanay's urban trees

Venue: Qostanai Smart City Hackathon · case No. 1

Role: Development and ML — the app and its cross-platform builds, model training, the inference service, the database and its access rules, offline sync

Stack: Vite · React 18 · TypeScript · PyTorch · MobileNetV3-Large · ONNX Runtime Web · Supabase · Leaflet · Capacitor 6 · Electron 33

- 343 photos in the dataset, split by tree 268 / 75
- 54,7% best validation accuracy
- 0,55 macro F1 across six classes
- 16,8 МБ ONNX model for CPU inference

QA Vision

2025–11 · 3 days

Paint-quality control on car bodies, from a photograph

Venue: Qostanai AI-Sana Industry Hackathon · Allur Challenge

Role: Development and architecture — the dashboard, the login service and API, the pluggable AI module system, the detection registry and KPI maths, the exports, roles and access control

Stack: Python · Streamlit · Plotly · FastAPI · Roboflow · Ultralytics YOLO · PyTorch · OpenCV · pandas · PyJWT · Power BI · Looker Studio

- 7 787 lines of Python in the project
- 4 kinds of AI module behind one interface
- 0,70 default confidence threshold
- 3 access roles: superadmin, plant manager, worker

KZ UniVerse

2025–12 · 24 hours

A single platform for Kazakhstan's universities

Venue: IT Fest 2025 · Hackathon track

Role: Development — the catalogue and university pages, comparison, analytics, the AI provider chain, profile and portfolio, the Telegram bot, and the demo rig for the pitch

Stack: Next.js 14 · TypeScript · Tailwind CSS · Framer Motion · Recharts · Ollama · Google Gemini 2.5 Flash Lite · Telegram Bot API

- 15 universities in the base
- 19 study programmes
- 3 assistant modes with fallbacks
- 24 ч for the whole track

NPAI

2025–12 · Since 2024, across several venues

A marketplace of ready-made industrial IoT modules

Venue: Startup Battle, IT Fest 2025 · top 30

Role: Product and development — the marketplace front, the dashboards, the Eco module, login and protected code preview, the project economics, the pitch and the answers to the jury

Stack: JavaScript · HTML/CSS · Google OAuth · IoT · Power BI

- 30 teams in the in-person Startup Battle final
- 3 cities where the project was shown
- 2024 the year of the first venue
- 4 revenue streams in the model

TIMELINE · 2024 → 2026

Idea battles and pitch sessions at Qostanai Hub

2024 · Kostanay

First outings with NPAI: learned to fit an idea into a short pitch, and to hear questions about money rather than code.

Pitch venue

2024–2025 · Petropavlovsk

A conversation with people from outside my own city, where the question of who actually buys this lands hardest.

Qostanai AI-Sana Industry Hackathon · Allur Challenge

2025 · Kostanay

1–3 November at Dulatov University. The first product built against a real manufacturing case — QA Vision.

IT Fest 2025 — two tracks

2025 · Almaty

5–6 December. A 24-hour hackathon with KZ UniVerse and Startup Battle with NPAI: top-30 teams, pitched in person.

Qostanai Smart City Hackathon · case No. 1

2026 · Kostanay

13–14 August. Jasyl: our own dataset, a fine-tuned model and a working offline app in two days.

Qostanai AgroTech Hackathon · 1st place

2026 · Kostanay

17–18 September. AgroVision AI: a weed map from drone imagery for Olzha Agro — first place on the case and a 500,000 ₸ prize.

Qostanai AI Industry Hackathon · final 2026 · Kostanay

October. НарядAI for Kostanay Minerals: top 4 in selection with 85 of 100 points, a final among 12 teams out of 150+ applications.

STACK

CONFIDENT	Python · JavaScript · HTML/CSS · FastAPI · aiogram · Git
USED IN PROJECTS	PyTorch · TensorFlow · MobileNetV2 · EfficientNet · YOLOv8 · Supabase · Leaflet.js · PWA · Render · Docker
LEARNING	React · n8n · Ollama · LM Studio

CERTIFICATES

- **ШАГ · School of Active Citizens** 2025–05
  Personal; the project was built with city budget funding
- **Pizza Pitch · Qostanai Hub** 2025–08
  Third place in the startup idea contest
- **Qostanai AI-Sana Industry Hackathon: Allur Challenge** 2025–11
  Personal; signed by Dulatov University, the ITU, Allur and Qostanai Hub
- **IT Fest 2025** 2025–12
  Personal; national festival, Almaty
- **Qostanai Smart City Hackathon** 2026–08
  Issued to the team, not to a person
- **Qostanai AgroTech Hackathon 2026 · победитель** 2026–09
  First place on the Olzha Agro case; issued to team MusorDropp
- **Qostanai AgroTech Hackathon 2026 · приз** 2026–09
  Cash prize for the winner — 500,000 ₸
- **Qostanai AgroTech Hackathon 2026 · участие** 2026–09
  Certificate of participation; issued to the team
- **Qostanai AI Industry Hackathon 2026 · участие** 2026–10
  Certificate of participation in the final; issued to the team

EDUCATION

- School-Lyceum No. 1, Kostanay — maths stream
- Music school — completed

LANGUAGES

Kazakh · Russian · English